

Bat guano contamination of karst spring water revealed by an automated microbial source tracking pipeline: Integrating amplicon sequencing and shotgun metagenomics

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ABSTRACT

Karst spring water in Korea is widely consumed untreated, yet its microbiological safety remains unexamined in bat-inhabited limestone cave regions. We developed MST-Pipeline, an open-source platform automating the complete microbial source tracking (MST) workflow from raw 16S rRNA amplicon data to source attribution, featuring automated V4 sub-region extraction for cross-region harmonization. Validation with fecal spike-in experiments confirmed dose-dependent source attribution, and PacBio long-read analysis demonstrated cross-platform compatibility. Application to the Mungyeong karst region (28 amplicon and 9 shotgun metagenomic samples) identified bat guano as the highest-contributing fecal source among the nine host groups tested in cave and stream water, with contributions higher in August (4.76%) than July (1.31%), coinciding with increased summer bat activity and antecedent rainfall. The downstream residential site showed distinct human and pig fecal signatures. Shotgun metagenomics revealed that guano harbors a wildlife-derived antibiotic resistance gene (ARG) reservoir dominated by multidrug efflux systems, while the downstream residential site carries anthropogenic ARG signatures including *sul1-qacEΔ1* integron cassettes at 4-fold higher abundance. Genome-resolved analysis identified contigs co-carrying ARGs and virulence factors corresponding to multidrug-resistant uropathogenic *Escherichia coli*, *Salmonella enterica* serovar Typhimurium, and *Shigella dysenteriae*. These findings demonstrate that karst spring water perceived as clean is contaminated with antibiotic-resistant pathogenic bacteria from bat guano, compounded by anthropogenic inputs downstream.

1. Introduction

Karst spring water has long been regarded as naturally pure in Korea. Over 1600 Community Drinking Water (CDW) facilities, locally known as "Yaksu," provide naturally flowing spring water for direct consumption, with surveys indicating that >20% of citizens in some regions rely on spring water as a primary drinking source (Kang et al., 2020). Many of these facilities are located in remote mountainous areas where regulatory oversight and water quality testing are limited, and the general

public assumes that geological filtration through fractured limestone ensures microbial safety.

However, this assumption is misleading. The same hydrogeological features that produce clear water—rapid infiltration through fractured limestone with minimal soil filtration—also facilitate rapid microbial transport, reducing pathogen retention time and allowing fecal microorganisms to move directly into aquifers (Sinreich et al., 2014). Karst groundwater systems are therefore highly vulnerable to biological contamination, particularly in environments inhabited by wildlife

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reservoirs such as bats.

Bats are well-recognized reservoirs of emerging zoonotic pathogens. While their role as viral reservoirs is extensively documented, including coronaviruses and paramyxoviruses (Calisher et al., 2006), the bacterial component of bat-associated microbiota remains comparatively understudied. Diverse pathogenic bacteria have been reported in bat guano, including *Salmonella*, *Escherichia*, *Campylobacter*, *Bartonella*, *Brucella*, and various Enterobacteriaceae with zoonotic potential (Ferreira et al., 2024). These bacteria can persist through environmental transmission routes, posing risks to both public health and agricultural safety. In Korea, karst regions provide ideal roosting habitats for large bat populations, and seasonal variation in bat activity—particularly during breeding and roosting periods in summer—may influence bacterial abundance in guano deposits and, consequently, contamination risk in adjacent water systems (Sakoui et al., 2020). Despite this, no systematic investigation has examined whether pathogenic bacteria originating from bat feces contaminate karst spring water that is directly consumed or used for agricultural irrigation.

Beyond which fecal sources reach drinking water, the genetic content they carry is a separate public-health concern. Antibiotic resistance genes (ARGs) and virulence factors (VFs) are increasingly recognized as molecular determinants of pathogen dangerousness, and water environments are a key conduit for their dissemination from animal and human gut microbiomes (Gillings et al., 2015; Haenelt et al., 2023). Two distinct ARG sources typically co-exist in human-influenced watersheds: a wildlife-derived background reservoir dominated by intrinsic efflux systems, and an anthropogenic input characterized by mobile genetic elements such as the class 1 integron-integrase gene *intI1* and its associated *suI1-qacEΔ1* cassette — widely used as molecular indicators of human wastewater and agricultural runoff (Gillings et al., 2015; Haenelt et al., 2023). ARGs and VFs co-localized on the same contig or plasmid are particularly clinically relevant, being characteristic of high-risk pathogenic lineages with enhanced environmental persistence and infection capability (Castelijn et al., 2012; Yaron and Romling, 2014; Zhuo et al., 2024). The intersection of wildlife fecal contamination, geological filtration, and anthropogenic inputs in karst spring water therefore represents a setting where ARG and VF surveillance is particularly informative for One Health risk assessment.

Microbial source tracking (MST) based on 16S rRNA gene amplicon sequencing provides a powerful culture-independent approach for attributing fecal contamination to specific host groups (Knights et al., 2011; Henry et al., 2016; Song and Unno, 2024). While qPCR-based MST using host-associated marker genes (e.g., HF183 for human, Bac-Cow for bovine) remains the more common operational approach in routine water-quality monitoring, sequencing-based community-wide MST offers broader source resolution and is increasingly adopted for multi-source environmental investigations (Unno et al., 2018). Moreover, a sequencing-based MST workflow requires researchers to manually integrate multiple bioinformatics tools—Cutadapt (Martin, 2011) for primer trimming, DADA2 (Callahan et al., 2016) for ASV inference, VSEARCH (Rognes et al., 2016) for sequence alignment, and SourceTracker2 (Knights et al., 2011) for source attribution—each with distinct installation requirements, input/output formats, and parameter settings. This fragmentation creates a steep learning curve, particularly for environmental scientists and public health professionals who may lack bioinformatics expertise (Boehm et al., 2013). Furthermore, different sequencing platforms and primer sets target different variable regions of the 16S rRNA gene, yet source tracking reference databases are typically constructed from a single region (commonly V4). Comparing sink samples amplified with V3-V4 or V4-V5 primers against a V4-only reference database requires careful sub-region extraction—a non-trivial step that is rarely automated.

Several additional knowledge gaps motivate this study. First, while MST techniques are well established for distinguishing human and livestock fecal sources, their application to wildlife-derived contamination—especially from bats—has been minimal (Choi et al.,

2025). Second, the occurrence of antibiotic resistance genes (ARGs) and virulence factors (VFs) in bat-derived bacteria in Korean karst ecosystems has never been examined. Third, no study has directly connected bat-associated bacterial contamination in karst spring water to the presence of clinically relevant antibiotic-resistant pathogens in waters used by human communities.

To address these challenges, we developed MST-Pipeline, an open-source, web-based platform that integrates the complete MST workflow into a single automated application with cross-region harmonization capabilities. In this study, we first validate the pipeline using controlled spike-in experiments and cross-platform sequencing data, then apply it alongside shotgun metagenomics to investigate bat guano contamination of karst spring water in the Mungyeong region of Korea. By integrating amplicon-based source tracking with functional metagenomic characterization of ARGs and VFs, we provide the first comprehensive assessment of how bat guano shapes the microbial safety of karst water systems consumed by rural communities.

2. Materials and methods

2.1. MST-pipeline: automated source tracking platform

MST-Pipeline is a web-based application built on FastAPI with a Plotly Dash interactive frontend and SQLite database backend (Fig. S1). The pipeline automates six sequential stages: (1) data upload with automatic detection of sequencing type (paired-end, single-end, PacBio long-read) and variable region (V4, V3-V4, V4-V5, V1-V9) detection via primer sequence matching with full IUPAC ambiguity support; when primer are not found, the pipeline aligns the input sequences to the *Escherichia coli* 16S rRNA gene using BMap (Bushnell et al., 2017) to identify the amplified region, and amplicons that do not cover V4 region are excluded from downstream analysis. To enable cross-region harmonization, MST-Pipeline accepts inputs from multiple primer sets (V4, V3-V4, V4-V5, V1-V9) and harmonizes all data to V4 by extracting the V4 sub-region from each ASV at the 515F (GTGY-CAGCMGCCGCGTAA) and 806R positions using regex matching with full IUPAC ambiguity. This allows V3-V4 or V4-V5 sink samples — the regions most widely used in environmental microbiome surveys — to be analyzed against a single V4 reference database without maintaining region-specific references. The auto-detection algorithm and its validation across V4, V3-V4, and V1-V9 amplicon datasets are described in detail in our companion paper (Unno, 2026). Accordingly, the pipeline supports V4, V3-V4, and V4-V5 Illumina amplicon data as well as V1-V9 long-read data. PacBio HiFi is the preferred long-read platform owing to its per-read accuracy sufficient for standalone genome assembly, whereas Oxford Nanopore Technology (ONT) reads may still require hybrid assembly to achieve comparable accuracy (Liang et al., 2025); (2) primer trimming using Cutadapt (Castelijn et al., 2012) with anchored 5' matching; (3) automated quality parameter optimization and ASV inference via DADA2 (Yaron and Romling, 2014), with platform-specific error models for Illumina and PacBio data; (4) taxonomy assignment against SILVA NR99 v138.1 (Quast et al., 2013) using the naive Bayesian classifier (Wang et al., 2007); (5) source tracking via SourceTracker2 with automated V4 sub-region extraction for cross-region harmonization; and (6) WHO-priority pathogen screening against 44 clinically relevant genera stratified by antimicrobial resistance priority. WHO-priority screening operates on amplicon-derived genus-level taxonomy for short-read data and species-level taxonomy for full-length long-read data.

The source reference database was constructed from 2168 samples spanning eleven host/source groups (bat, human, pig, cow, horse, groundwater, chicken, duck, bird, seawater, river). The bat reference (n = 515; BioProject PRJNA1436137; Unno et al., in review) is the fecal subset of a larger 2747-sample Korean bat-microbiome survey; the wild-bird reference (n = 68; BioProject PRJNA981578; Youngblut et al. (Youngblut et al., 2021)) spans 12 species across 9 avian orders. Other

livestock and environmental sources are obtained from our previous study (Song and Unno, 2024), collected between 2016 and 2020 from poultry farms, cow farms, swine farms, horse stables, and human donors across Jeju and Gwangju, South Korea. Fecal samples were stored at -20°C until DNA extraction with the PowerFecal Isolation Kit (MoBio, Carlsbad, CA, USA). The 16S rRNA V4 region was amplified with primers 515F/806R, or the V3-V4 region with primers 341F/805R, following Illumina's 16S Metagenomic Sequencing Library Preparation protocol, and sequenced on the Illumina MiSeq platform (Macrogen, Seoul, South Korea). All datasets were independently processed through the DADA2 pipeline and harmonized to the V4 region.

After two-stage rare-ASV filtering (prevalence ≥ 2 ; relative abundance $\geq 0.05\%$) and per-group Bray-Curtis outlier detection (median $+ 3 \times \text{MAD}$; 84 outliers, per-group counts in Table S1), the quality-controlled database contained 2084 samples and 37,622 ASVs. PCoA and pairwise PERMANOVA confirmed clear separation of the seven fecal groups (21 pairs, $R^2 = 0.04\text{--}0.29$, all $p_{\text{adj}} < 0.001$; Fig. S2). Full details of sample provenance, sequencing platforms, V4 extraction, outlier QC, and pairwise validation are provided in Supplementary Methods S1. For the karst water application, horse and seawater source groups were excluded as ecologically irrelevant to the inland study site, yielding a final analysis database of nine source groups (bat, bird, chicken, cow, duck, groundwater, human, pig, river; 1810 samples). MST-Pipeline is freely available at <https://github.com/tatsul207/MST-Pipeline> under the MIT License; the source reference database is bundled with the release.

2.2. Pipeline validation: spike-in experiment

To validate MST-Pipeline's source attribution accuracy, we performed controlled spike-in experiments in which cow or pig fecal slurry was spiked into 300 mL river water at two concentrations (1% and 5% vol/vol), with three biological replicates per condition. Fecal slurries were prepared as described previously (Choi and Unno, 2026): feces were collected from five cows and five pigs (a local cattle farm and a local swine farm, Cheongju, Korea); 2 g of feces from each of the five individuals per species was homogenized into a single composite sample to average inter-individual gut-microbiome variation; 10 g of the composite was suspended in 300 mL of sterile phosphate-buffered saline (PBS, pH 7.4) with glass beads; and the suspension was passed through a 5.0- μm mixed-cellulose-ester membrane to remove large particles. The resulting filtrate was used as the fecal slurry. As positive controls, the same cow and pig composite fecal samples were also sequenced directly, without dilution into river water (i.e., neat feces, with no organism added); these "unspiked" controls verify the single-source attribution expected for each species. All 14 samples were sequenced on the Illumina MiSeq platform using V3-V4 primers (341F/805R) generating 300 bp paired-end reads (BioProject PRJNA1417490). Additionally, three PacBio HiFi full-length 16S rRNA samples (SRA accessions SRR23380902–SRR23380904) representing human fecal microbiomes from an Italian periodontitis cohort (Baima et al. (Baima et al., 2024)) were processed to validate cross-platform V4 harmonization. The V3-V4 and PacBio datasets were combined in a single source-tracking analysis to demonstrate multi-region integration.

For cross-region generalizability testing, we additionally analyzed the Australian fecal source library from Lim et al. (Lim et al., 2025) (BioProject PRJNA309092; Melbourne, V3-V4 Illumina MiSeq) as sinks against the unmodified 11-group Korean reference. In-silico mock spike-ins were constructed by computationally mixing Australian cow, chicken, and human reads with Korean river-water reads at 0–100% spike levels (30,000 reads per mock; three random-seed replicates per level; 45 mocks total, Table S2).

2.3. Study site and sample collection: karst water investigation

Samples were collected from cave-associated environments in the

Mungyeong karst area, Korea, during two sampling campaigns in July and August 2025. The study area encompasses two limestone caves—Sutgul and Amgul—and a surface stream system that flows from the cave exits through a small residential area (Fig. 1).

A total of 28 samples were collected for 16S rRNA amplicon sequencing: 16 water samples (8 per month) from three spatial clusters along the water flow path (Fig. 1). For each water sample, 12 L was collected in sterile containers, prefiltered through 5.0- μm mixed cellulose ester (MCE) membrane filters (47-mm diameter; Advantec, Japan) to remove debris, and then filtered through 0.22- μm MCE membrane filters (47-mm diameter; Advantec, Japan). Filters were stored at -80°C until DNA extraction. The Sutgul cluster comprised W1–W2 (cave interior), W3 (cave entrance), W4 (cave mouth), and W7 (adjacent spring), all located within approximately 30 m of each other. The Amgul cluster comprised W5 (cave interior) and W6 (cave entrance), approximately 200 m west of Sutgul. W8 was located approximately 600 m downstream at a stream entering the residential area. Additionally, 12 guano samples (6 per month) were collected from guano deposits inside both caves, including entrance and interior sampling points to capture spatial variation.

2.4. DNA extraction and sequencing

DNA was extracted from approximately 0.25 g of guano or 250 mL equivalent of filtered water using the DNeasy PowerSoil Pro Kit (Qiagen, USA) following the manufacturer's protocol. For amplicon sequencing, the V3-V4 region of the 16S rRNA gene was amplified using primers 341F (5'-CCTACGGGNGGCWGCAG-3') and 805R (5'-GACTACHVGGG-TATCTAATCC-3'). PCR amplification was performed in triplicate 25 μL reactions containing 12.5 μL KAPA HiFi HotStart ReadyMix, 0.2 μM of each primer, and 10 ng of template DNA. Paired-end sequencing (2×300 bp) was performed on the Illumina MiSeq platform.

For shotgun metagenomics, 9 libraries were prepared: guano DNA from Sutgul (F1) and Amgul (F2) caves was pooled separately to generate two composite libraries, while each water sample (W1, W3–W8) was processed individually to preserve site-specific variation. Metagenomic libraries were sequenced on the Illumina NovaSeq 6000 platform (2×150 bp paired-end mode), generating 84–106 million reads per sample.

2.5. Amplicon data processing and source tracking

Raw paired-end reads were processed using MST-Pipeline with automatic V3-V4 region detection, primer trimming, quality optimization, ASV inference, chimera removal, and taxonomy assignment against SILVA v138.1. The pipeline yielded 24,885 ASVs across 28 samples with an overall 56.9% read retention rate through the full DADA2 workflow. Alpha-diversity indices (Chao1 richness and Shannon diversity) were calculated using the *vegan* package in R. Statistical comparisons between groups (taxon abundances, ARG/VF abundances, and source attribution proportions) were conducted using the Wilcoxon rank-sum test with Benjamini-Hochberg FDR correction. Beta-diversity was quantified using Bray-Curtis dissimilarity, visualized with principal coordinates analysis (PCoA), and tested with PERMANOVA.

For source tracking, the V4 sub-region was extracted from each V3-V4 ASV by locating the 515F primer sequence within each ASV using fuzzy regex matching, retaining the downstream sequence. This yielded 14,099 unique V4 ASVs from 24,885 input ASVs (17,219 primer matches, 69.2%). V4-extracted sink ASVs were aligned against the 11-group source reference database using VSEARCH at 99% identity. Source proportions were estimated using SourceTracker2's Gibbs sampling algorithm with 10 restarts, 100 burn-in iterations, and rarefaction depth of 1000 for both sources and sinks. Two analyses were performed: (1) all nine source groups (after exclusion of horse and seawater as ecologically irrelevant) to assess environmental and fecal contributions, and (2) fecal sources only (bat, bird, chicken, cow, duck, human, pig) to



Fig. 1. Satellite image of the study site in the Mungyeong karst area, Korea, showing three sampling clusters. The Sutgul cluster (W1–W4, W7) includes cave interior, entrance, mouth, and adjacent spring sites within ~30 m. The Amgul cluster (W5, W6) includes cave interior and entrance sites ~200 m west. W8 is located ~600 m downstream at a stream entering the residential area. Guano samples (F1, F2) were collected from deposits inside Sutgul and Amgul caves, respectively.

focus on fecal attribution. Results are presented excluding the Unknown fraction to emphasize relative contributions among identified sources. The values reported by SourceTracker2 are mixing proportions estimated by Bayesian Gibbs sampling — the estimated proportion of the sink community, expressed as a fraction of its 16S rRNA gene sequence reads, that is attributable to each source group; that is, a read-based relative abundance (a cellular fraction of the sequenced community), and not a volumetric fraction of the original water sample or an absolute biomass fraction. Because such 16S read fractions are additionally shaped by inter-taxon variation in 16S rRNA gene copy number and by the total microbial load of the sink, they are interpreted here as comparative (ordinal) indicators across sites, months, and host groups rather than as absolute contamination fractions (Choi and Unno, 2026).

2.6. Shotgun metagenomic analysis

Raw shotgun metagenomic reads were subjected to quality control using fastp (v0.23.4) (Chen et al., 2018) with removal of adapter sequences, reads with average Phred quality scores < 30, and reads shorter than 50 bp. Read retention after QC ranged from 94.3% to 97.5% across samples. Quality-filtered reads were assembled *de novo* using MEGAHIT (v1.2.9) (Li et al., 2016) with k-mer sizes ranging from 27 to 141 and minimum contig length of 1000 bp, yielding 29,483–81,036 contigs per sample with N50 values of 1834–4380 bp.

ARGs and VFs were quantified using a read-based approach by aligning quality-filtered reads to the Comprehensive Antibiotic Resistance Database (CARD v3.2.9) (Alcock et al., 2023) and the Virulence Factor Database (VFDB, August 2024 release) (Zhou et al., 2025) using KMA (v1.4.11) (Clausen et al., 2018). Gene abundance was normalized

as reads per kilobase per million mapped reads (RPKM). For assembly-based gene discovery, open reading frames (ORFs) were predicted from assembled contigs using Prodigal (v2.6.3) (Hyatt et al., 2010) in metagenomic mode. Predicted ORFs were annotated for ARGs and VFs using ABRicate (v1.0.1) (<https://github.com/tseemann/ABRICATE>) against CARD and VFDB with minimum identity of 90% and minimum coverage of 80%. Contigs carrying both ARG and VF annotations were identified as potential high-risk pathogenic elements, and their taxonomic affiliation was inferred using MMseqs2 (Steinberger and Soding, 2017) against the SWISS-PROT database. Plasmid mobility

was assessed using MOB-typer (Robertson and Nash, 2018) and geNomad (Camargo et al., 2024) to evaluate the potential for horizontal gene transfer.

3. Results

3.1. Pipeline validation

MST-Pipeline automatically detected the V3-V4 variable region (87% primer match confidence), trimmed primers with Cutadapt, and

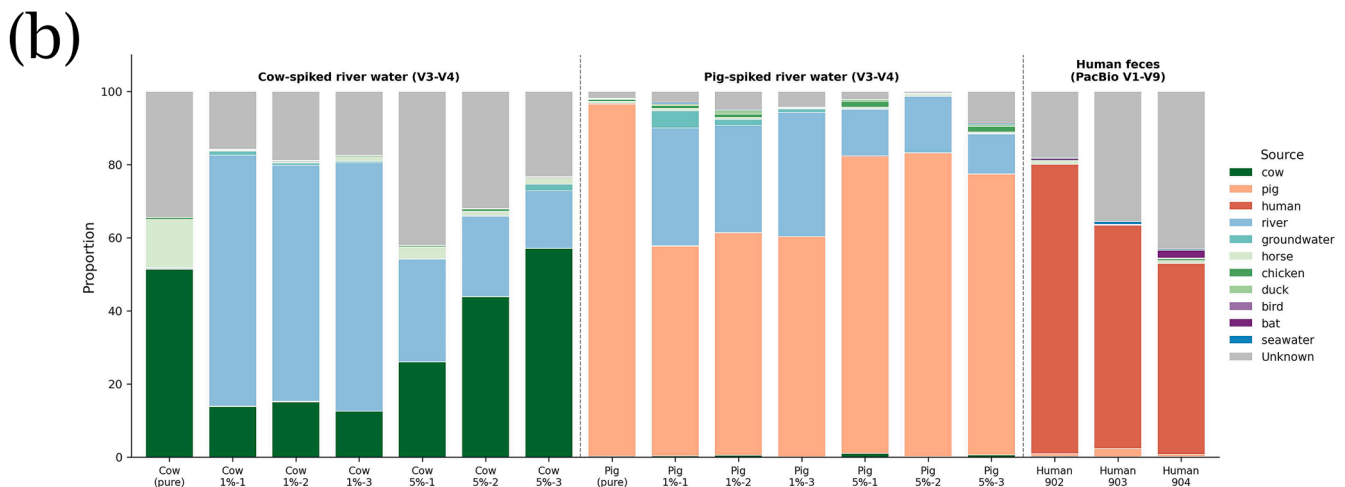
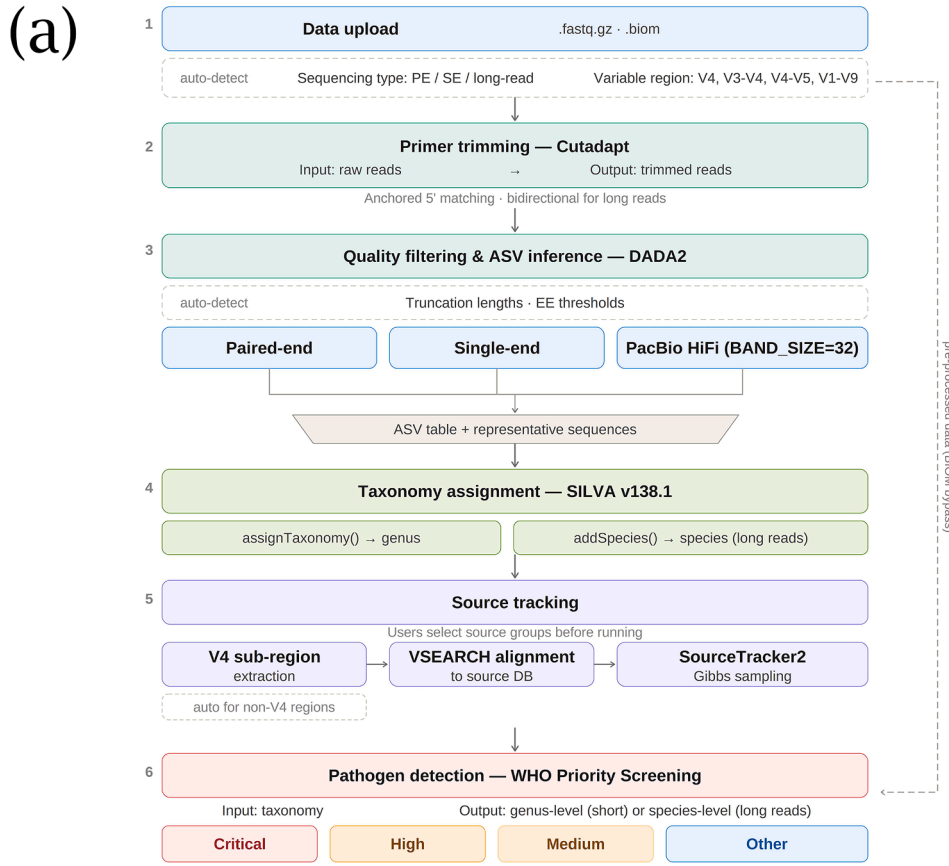


Fig. 2. MST-Pipeline validation. (A) Pipeline workflow showing automated sequencing detection, primer trimming, quality optimization, ASV inference, V4 extraction, and source tracking. (B) Source tracking results for the spike-in experiment showing dose-dependent attribution of cow and pig fecal slurry spiked at 1% and 5% (vol/vol) into river water, and cross-platform validation with PacBio long-read human fecal samples correctly attributed to human source after automatic V4 sub-region extraction. Attribution percentages are read-based fractions of 16S rRNA gene sequence reads assigned to each source group.

optimized truncation parameters (trunc_len_f = 284, trunc_len_r = 279) for the spike-in samples. DADA2 processing yielded 336 ASVs across 14 samples after chimera removal. Because the sink samples were sequenced with V3-V4 primers while the source reference database is V4-based, MST-Pipeline automatically extracted the V4 sub-region from each ASV, yielding 205 unique V4 ASVs from 336 input ASVs. Feature alignment against the source database matched 378 of 468 combined ASVs (80.8%), with per-sample read retention of 91–96%.

Source tracking demonstrated accurate, dose-dependent attribution. Pure pig feces were attributed primarily to pig (93.5%) with low Unknown (4.2%), and pure cow feces were attributed primarily to cow (34.0%) with the cross-reactive ruminant signal absorbed into Unknown. The cow attribution is lower than would be obtained when horse is included as a candidate source group (in which case 51.4% cow + 13.2% horse = 64.6% ruminant-associated, reflecting shared ruminant gut-microbiome features). For spike-in samples, a clear dose-response relationship was observed: cow attribution increased from $4.2 \pm 0.5\%$ at 1% to $20.4 \pm 18.0\%$ at 5%, while pig attribution increased from $54.3 \pm 1.8\%$ at 1% to $74.4 \pm 4.9\%$ at 5%. No spurious attributions to ecologically implausible sources (e.g., bat, seawater) were observed, and pathogen detection identified *Clostridium* spp. in cow-associated samples, consistent with known bovine gut microbiota (Holman and Gzyl, 2019).

For cross-platform validation, MST-Pipeline automatically detected the V1-V9 region in PacBio samples, applied PacBio-specific DADA2 parameters (PacBioErrfun error model, BAND_SIZE = 32, length filter 1000–1600 bp), and extracted the V4 sub-region from 676 full-length ASVs, producing 269 unique V4 ASVs. All three PacBio samples were correctly attributed to human as the dominant source (52.2–79.2%), with remaining proportions classified as Unknown (18.1–43.2%). The combined analysis of V3-V4 and V1-V9 samples in a single source tracking run demonstrated successful cross-platform harmonization (Fig. 2).

To test cross-region generalizability, we additionally analyzed Australian fecal samples from BioProject PRJNA309092 (Lim et al., 2025) as sinks against the Korean reference used in this study. The database correctly identified the dominant source for cow (10/10, mean 79.8%) and horse (7/7, mean 90.0%), with moderate accuracy for chicken (5/8 dominant correct, mean 53.5%); the remaining three showed expected cross-reactivity with duck. Hosts absent from the database cross-attributed in biologically meaningful patterns (deer→cow 61.6%; wombat→horse 65.8%; kangaroo→cow 49.6%; possum→Unknown 83.4%), reflecting phylogenetic convergence rather than arbitrary noise. In-silico mock spike-ins (Australian cow/chicken/human reads computationally mixed with river water at 0–100%) showed monotonic dose-dependent attribution for cow and chicken, with no false-positive fecal attribution in pure-water controls. Detailed Australian validation results are provided in Table S2.

3.2. Microbial community composition of karst water and guano

Guano and water samples harbored distinct microbial communities. Water samples exhibited significantly higher alpha-diversity than guano (Chao1: adjusted $p = 6.6 \times 10^{-8}$; Shannon: adjusted $p = 4.6 \times 10^{-7}$), indicating that aquatic microbial communities are more taxonomically rich and even, whereas guano microbiota are dominated by a smaller number of taxa. PCoA based on Bray-Curtis dissimilarity confirmed clear separation between sample types along PC1 (17.1% variance explained) and PC2 (11.7%), with PERMANOVA confirming statistically significant community differences ($F = 12.47$, $R^2 = 0.32$, $p < 0.001$; Supplementary Fig. S3).

At the family level, guano samples were dominated by Enterobacteriaceae, Streptococcaceae, and Clostridiaceae, while water samples exhibited higher relative abundances of Flavobacteriaceae, Burkholderiaceae, and Sphingomonadaceae. Indicator species analysis identified nine genera significantly enriched in guano (FDR-adjusted $p <$

0.05), led by *Lactococcus* (9.9% mean relative abundance in guano vs. 0.2% in water), *Enterobacter* (9.6% vs. 0.3%), and *Enterococcus* (3.8% vs. 0.3%), reflecting their strong association with intestinal environments (Supplementary Fig. S4). Other feces-associated genera including *Candidatus Arthromitus*, *Serratia*, *Rhodococcus*, *Solibacillus*, and *Arthrobacter* were also significantly more abundant in guano.

3.3. Source attribution in karst water

As an additional validation, guano samples collected from Sutgul and Amgul caves in 2025 were run as sinks against the source database. Bat showed the highest contribution compared to other fecal sources in all 12 guano samples, confirming the discriminatory power of the reference database for bat fecal microbiomes (Table S3). Two guano samples (F_JUL_04, F_AUG_02), however, showed lower bat attribution (6.4–8.1% bat attribution, yet highest among sources), possibly reflecting degraded or environmentally weathered guano deposits (Table S3). F_JUL_04 also had the highest bird attribution among all 12 guano samples (6.8%), so minor avian contact at the deposit surface cannot be fully excluded (Table S3).

Source tracking of the 16 water samples revealed distinct patterns depending on the source groups included (Fig. 3). When all nine source groups were analyzed, groundwater and river were the dominant identified contributors (Fig. 3A), consistent with the karst hydrological setting where infiltrating groundwater and surface stream flow constitute the primary water sources. Among the fecal source groups, however, bat emerged as the largest contributor — non-zero in all 16 water samples and consistently higher in August than July (Fig. 3B).

The downstream residential site W8 showed a distinct fecal profile: human (5.9% in August; 2.2% in July) and pig (1.6%/0.7%) were the dominant fecal sources, consistent with anthropogenic input from the nearby community across both sampling months. No other water sample showed substantial human or livestock contributions (all <2%), confirming that bat guano is the primary fecal contaminant in the cave-associated water system.

Temporal analysis of bat-associated opportunistic pathogens in water samples revealed that *Enterobacter* — a genus significantly enriched in guano — was significantly more abundant in August than July (Wilcoxon $p = 0.005$), consistent with the higher August guano deposition driving pathogen transfer to the aquatic environment (Fig. S5). Other shared genera including *Escherichia-Shigella*, *Providencia*, *Mycobacterium*, and *Vibrio* were detected in both guano and water samples, indicating possible transmission of bat-associated bacteria into the aquatic environment.

3.4. Resistome and virulome characterization

ARGs were detected in all samples (Table 1), with total RPKM values of 654,001–815,591 in guano and 570,483–1091,911 in water. Notably, W8 (residential stream) exhibited the highest total ARG abundance (1091,911 RPKM), nearly twice the level observed at upstream cave sites (W5: 570,483; W6: 577,604). Hierarchical clustering of the top 30 most abundant ARGs revealed two distinct profiles (Fig. 4A).

Guano samples were enriched (FDR-adjusted $p < 0.05$) in multidrug efflux pump regulators (*gadX/W*, *ramA*, *mdtE/F*) and fluoroquinolone resistance genes, representing a resistance signature intrinsic to bat gut-associated microbiota and likely reflecting the natural resistome rather than selective pressure from clinical antibiotic use. In contrast, water samples — particularly W8 — displayed elevated aminoglycoside resistance genes (APH(3')-Ic, APH(6)-Id, AAC(6')) and the *sul1-qacEΔ1* integron cassette, a well-established marker of class 1 integrons and anthropogenic contamination. At W8, *sul1* abundance reached 211,926 RPKM and *qacEΔ1* reached 146,085 RPKM, 4–5 fold higher than any other site (Fig. 4B), consistent with input from wastewater or agricultural runoff associated with the adjacent residential area.

Analysis by antibiotic class revealed that aminoglycoside,

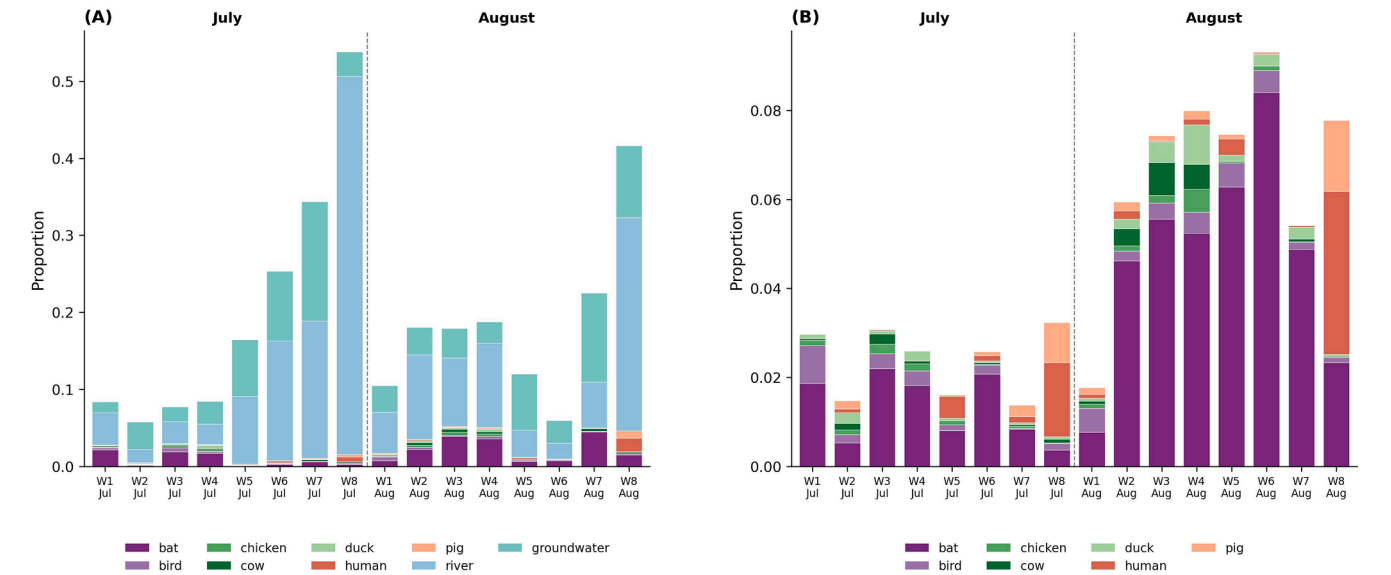


Fig. 3. Microbial source tracking results for karst water samples (Unknown fraction excluded to show actual source proportions). (A) Source contributions from all nine source groups (horse and seawater excluded as ecologically irrelevant; bat, bird, chicken, cow, duck, groundwater, human, pig, river), showing groundwater and river as the dominant identified contributors. (B) Fecal source contributions only (bat, bird, chicken, cow, duck, human, pig), showing bat guano as the highest-contributing fecal source among the tested host groups, with higher contributions in August than July and a distinct human/pig signature at the downstream residential site W8. Dashed lines separate sampling months. Reported source proportions are read-based fractions (the estimated percentage of 16S rRNA gene sequence reads assigned to each source group), not volumetric or cell-count fractions.

Table 1
Total abundance of antibiotic resistance genes (ARGs) and virulence factors (VFs) in guano and karst water samples (KMA-RPKM).

Sample	Type	Location	ARG (RPKM)	VF (RPKM)
F1	Guano	Sutgul cave	815,591	888,852
F2	Guano	Amgul cave	654,001	854,051
W1	Water	Sutgul cave interior	779,201	819,719
W3	Water	Sutgul cave entrance	716,799	820,054
W4	Water	Sutgul cave mouth	740,399	827,845
W5	Water	Amgul cave interior	570,483	866,913
W6	Water	Amgul cave entrance	577,604	865,361
W7	Water	Adjacent spring	715,713	883,129
W8	Water	Residential stream	1091,911	905,017

fluoroquinolone, and tetracycline resistance genes were the most abundant across both sample types, while sulfonamide and diaminopyrimidine resistance showed marked elevation at downstream water sites (Fig. 4B).

VF abundance was uniformly high across all samples (819,719–905,017 RPKM), but functional composition differed substantially between guano and water (Fig. 5). Water samples were strongly enriched in environmentally adaptive virulence traits: flagella-associated motility genes (31–43% of total VF abundance), Type IV pili for adherence (17–24%), alginate biosynthesis for biofilm formation (3–15%), and enterobactin for iron acquisition. In contrast, guano sample F2 showed a distinctive enteric pathogen signature enriched in exotoxins (18%), iron uptake/metabolism (29%), and adherence factors (29%) with minimal motility genes (3%), reflecting the virulence repertoire of enteric pathogens adapted to the gut environment rather than free-living aquatic conditions.

Pathogen inference from VF signatures identified *Pseudomonas aeruginosa* as the dominant VF-associated pathogen in water samples, particularly at downstream sites W7 and W8, consistent with its ecology as an environmental opportunist. Guano sample F2 harbored a distinct enteric pathogen profile: UPEC, *Yersinia pestis*, *Shigella dysenteriae*, *Salmonella enterica* serovar Typhimurium, *Enterobacter cloacae*, and *Klebsiella pneumoniae* (Fig. S6).

3.5. Co-localized ARGs and virulence factors

Genome-resolved analysis identified five contigs simultaneously carrying ARGs and VFs, representing potentially high-risk pathogenic bacteria with antibiotic resistance capabilities (Table 2). Guano sample F2 yielded four contigs: (1) multidrug-resistant UPEC with efflux systems (*gadX/W*, *mdtE/F*) and the complete *chu* heme uptake operon (*chuA* through *chuV*); (2) UPEC with efflux regulator (*ramA*) and enterobactin biosynthesis and transport genes (*entB*, *entS*, *fepG*, *fepC*); (3) multidrug-resistant *S. enterica* Typhimurium with efflux pumps (*mdtG/H*) and curli fimbriae for biofilm formation (*csgG/F/D/B*); and (4) *S. dysenteriae* with outer membrane efflux protein (*tolC*) and a complete Type II secretion system (*gspC* through *gspM*). Water sample W1 yielded a contig carrying *MexB-OprM* multidrug efflux system genes together with Type IV pilus assembly genes (*pilG* through *pilJ*), confirming the presence of multidrug-resistant *P. aeruginosa* with motility and adherence capabilities in the cave stream environment.

The co-localization of resistance and virulence determinants on the same contigs is characteristic of high-risk pathogenic lineages (Castelijn et al., 2012; Yaron and Romling, 2014; Zhuo et al., 2024) with enhanced environmental persistence and infection potential. Plasmid mobility analysis showed that most ARG/VF-carrying contigs were non-mobilizable (chromosomally encoded), suggesting stable vertical inheritance. However, one mobilizable contig from guano (k141_319834) carried macrolide resistance genes (*mphE*, *msrE*) with a MOBQ relaxase, indicating that horizontally transferable resistance elements exist within the bat guano microbiome. Downstream water samples (W3, W4, W8) carried ARG-only plasmid contigs with sulfonamide, aminoglycoside, and tetracycline resistance, consistent with anthropogenic contamination sources (Table S4).

4. Discussion

4.1. Validated pipeline enables reliable source attribution

Sequencing-based community-wide MST has accumulated a substantial validation literature. The Bayesian Gibbs-sampling approach implemented in SourceTracker was originally benchmarked against

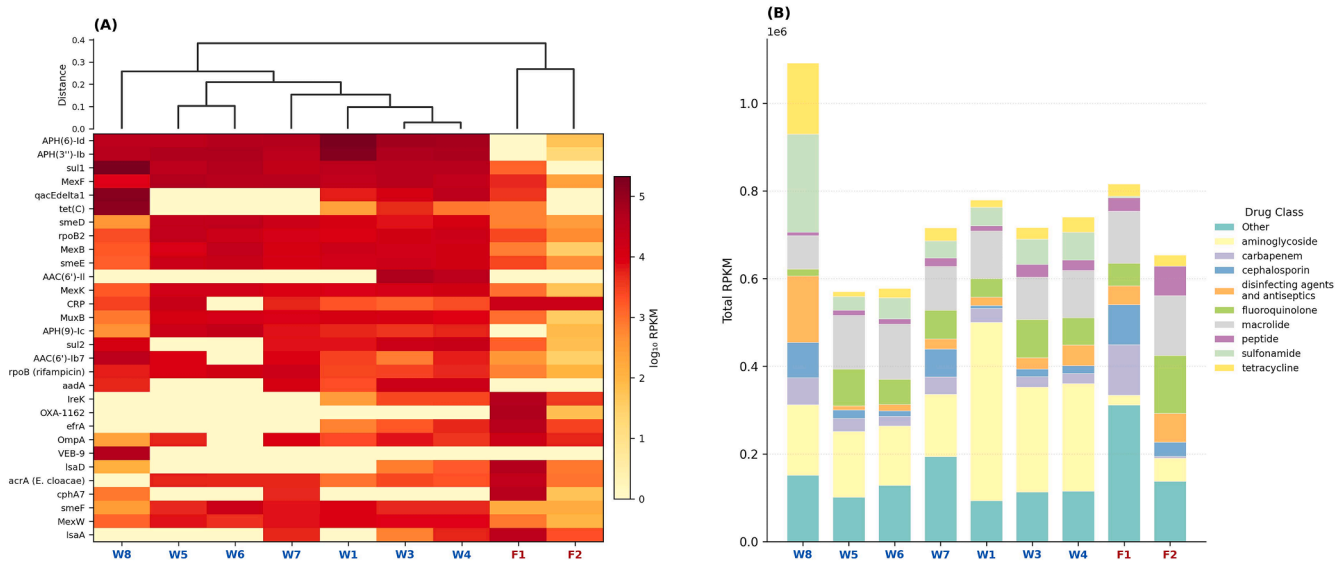


Fig. 4. Antibiotic resistance gene profiles in guano and water. (A) Heatmap of the top 30 most abundant ARGs ($\log_{10}$ RPKM) with hierarchical clustering separating guano and water samples. (B) Total ARG abundance by antibiotic drug class per sample, showing wildlife-derived signatures in guano and anthropogenic signatures (aminoglycoside, sulfonamide) elevated at downstream water sites, particularly W8.

Virulence Factor Abundance: Guano vs. Water Samples

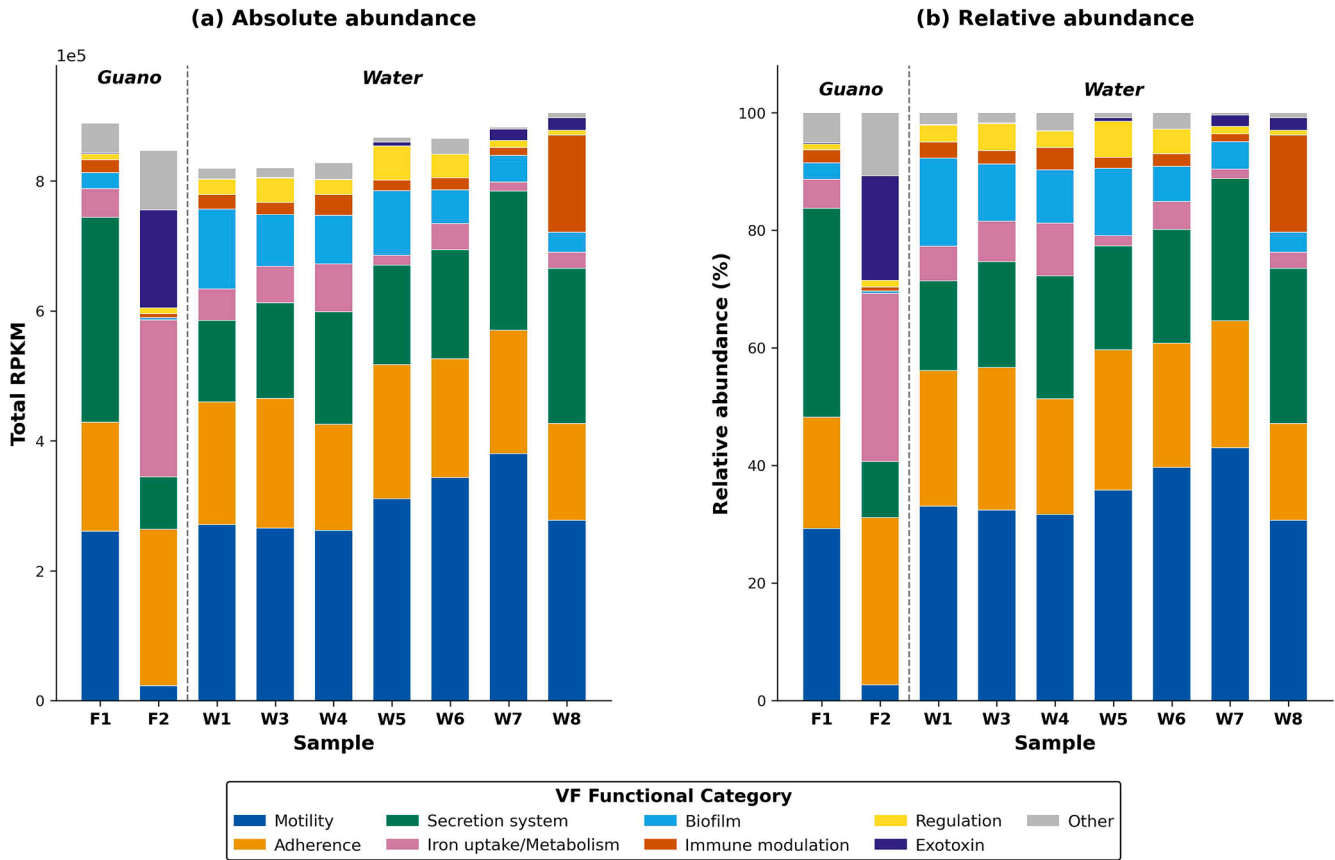


Fig. 5. Virulence factor abundance in guano and water samples by functional category. (A) Absolute abundance (total RPKM) and (B) relative abundance (%) showing enrichment of motility, biofilm, and iron acquisition genes in water versus exotoxin and enteric-associated virulence in guano, particularly F2.

simulated and real-data mixtures (Knights et al., 2011), extended to coastal-water fecal contamination by Henry et al. (Henry et al., 2016), and assessed in a 27-laboratory comparison of 41 MST methods by Boehm et al. (Boehm et al., 2013). Staley et al. (Staley et al., 2018)

reported a double-blinded validation in which SourceTracker correctly identified the contamination source in 31 of 34 spiked freshwater samples (91% accuracy) with monotonic dose-response (Spearman $\rho \geq 0.88$ across 0.025–10% spike volumes), and the fecal-source-library

Table 2

Contigs co-carrying antibiotic resistance genes and virulence factors from metagenomic assemblies.

Sample	ARGs	Virulence Factors	Inferred Pathogen
F2 (guano)	gadX, gadW, mdtE, mdtF	chuA-V (heme uptake)	MDR <i>E. coli</i> (UPEC)
F2 (guano)	ramA	entB, entS, fepG, fepC (enterobactin)	MDR <i>E. coli</i> (UTI)
F2 (guano)	mdtG, mdtH	csgG, csgF, csgD, csgB (curli fimbriae)	MDR <i>S. enterica</i> Typhimurium
F2 (guano)	tolC	gspC-M (Type II secretion)	MDR <i>S. dysenteriae</i>
W1 (water)	MexB, OprM	pilG-J (Type IV pilus assembly)	MDR <i>P. aeruginosa</i>

validation framework of Lim et al. (Lim et al., 2025) provides quantitative leave-one-out specificity metrics for source databases of this type. Our V4-harmonized spike-in validation contributes a further dose-response example (cow 4.2 → 20.4%; pig 54.3 → 74.4% across 1–5% spikes) to this growing evidence base.

Beyond this primary spike-in validation, three additional strands reinforce the pipeline's reliability for environmental application: (i) cross-platform PacBio long-read data from a non-Korean (Italian) human cohort were correctly attributed to human at 52–79%, providing an external geographic check on the Korean-built reference; (ii) the Australian generalizability test (Table S2) showed strong cross-region transfer for cow (10/10, mean 79.8%) and horse (7/7, mean 90.0%), with biologically meaningful cross-attribution for hosts absent from the database, indicating that attribution is driven by phylogenetic microbiome similarity rather than noise; and (iii) the 12 study-site guano deposits were correctly attributed to bat in 10/12 (Table S3). The source reference database itself was quality-controlled through per-group Bray-Curtis outlier detection; pairwise PERMANOVA confirmed that the seven fecal groups formed distinct clusters ($R^2 = 0.04$ – 0.29 , all $p_{\text{adj}} < 0.001$; Fig. S2), supporting the discriminatory power of the database for multi-source attribution. However, absolute proportions estimated by SourceTracker2 are not strictly quantitative measures of contamination fraction: the Bayesian Gibbs sampling estimates proportions relative to a finite source set, and phylogenetically related hosts produce expected cross-attribution. For the karst water analysis we therefore interpret source proportions as ordinal/comparative indicators (site-to-site, month-to-month, across-host-group) rather than absolute fractions of the sink microbial community. Concretely, these proportions are fractions of 16S rRNA gene sequence reads assigned to each source group — a read-based relative abundance, i.e., a cellular fraction of the sequenced community — not cell counts or volumetric fractions of the sampled water; because read fractions are sensitive to among-taxon 16S gene-copy-number differences and to sink microbial biomass, both of which can shift inferred source proportions even at fixed spike-in levels (Choi and Unno, 2026), cross-study comparison of absolute percentages should be made with caution.

Compared to existing platforms, MST-Pipeline distinguishes itself through integration scope. QIIME 2 (Bolyen et al., 2019) provides modular amplicon analysis but requires command-line expertise and does not include integrated source tracking. SourceTracker2 performs source attribution but does not handle upstream data processing. No existing platform combines automated primer detection, quality optimization, cross-region harmonization, source tracking, and pathogen detection in a single web interface accessible to non-bioinformaticians. The automated quality parameter selection — which eliminates the need for manual inspection of quality profiles — is particularly valuable for water quality monitoring programs where standardized, reproducible analysis is essential.

4.2. Bat guano as a major summer-period fecal source in karst water

MST analysis identified bat guano as the highest-contributing fecal source detected in cave and stream water across the Mungyeong karst system among the nine tested host groups. Bat consistently contributed the largest fecal-source proportions in both sampling months (mean

4.76% in August vs. 1.31% in July), while all other livestock and human sources were negligible (<2%) except at the downstream residential site W8, where distinct anthropogenic fecal signatures were detected in both months (human 5.9% and pig 1.6% in August; human 2.2% and pig 0.7% in July). The increase in bat fecal contributions from July to August likely reflects a combination of breeding-season dynamics and antecedent rainfall patterns.

In this study, direct bat counts were not collected, but Korean cave-bat ecology provides relevant context: in limestone caves serving as active-season maternity or roost sites, bat populations and per-cave fecal output peak during the summer breeding period (May–August), typically reaching hundreds to several thousand individuals per cave and up to ~15,000 in major *Miniopterus fuliginosus* maternity colonies (Kim et al., 2014). Small non-bat cave fauna (rodents, amphibians) are typically rare in temperate Korean limestone caves and contribute orders of magnitude less bacterial load than bat colonies; any minor contribution is subsumed into the Unknown fraction, as these taxa are not represented as discrete source groups.

Precipitation data from the Korea Meteorological Administration (Fig. S7) revealed that July sampling (7/28) occurred after a 6-day dry period following heavy rainfall (132 mm over July 16–17), whereas August sampling (8/11) occurred only 2 days after the most recent rainfall (1.4 mm on 8/9), following moderate rainfall earlier in the month (39 mm over August 3–4). More recent rainfall prior to August sampling would have actively mobilized fresh guano deposits into the cave stream system, while the extended dry period before July sampling allowed fecal inputs to attenuate. Combined with increased bat activity during the breeding season (Sakoui et al., 2020), these hydrological dynamics suggest that both biological (bat activity) and meteorological (rainfall-driven flushing) factors drive the temporal variation in bat fecal contamination. This has practical implications: water quality monitoring programs in karst regions should incorporate both seasonal and rainfall-event considerations, with heightened surveillance during summer months when bat activity and monsoon rainfall coincide.

The high proportion of Unknown in water samples (87–99%) reflects the dominance of environmental bacteria not represented in any fecal source database—an expected and well-documented phenomenon for karst groundwater systems (Henry et al., 2016). This does not diminish the significance of the bat fecal signal; rather, it indicates that while fecal contamination represents a small fraction of the total microbial community, it carries disproportionate public health significance through the introduction of enteric pathogens and ARGs. The taxonomic data support this interpretation: indicator species analysis confirmed that enteric genera including *Enterobacter*, *Lactococcus*, and *Enterococcus*—which are significantly enriched in guano—are also detectable in water samples, providing independent evidence of fecal-aquatic microbial transfer.

4.3. Dual ARG sources in the karst watershed

A critical finding of this study is the evidence for two distinct ARG sources operating within the same watershed. The wildlife-derived ARG reservoir in guano is characterized by multidrug efflux pump regulators (*gadX/W*, *ramA*, *mdtE/F*) and fluoroquinolone resistance genes, representing a resistance signature intrinsic to enteric pathogens (UPEC,

Salmonella, *Shigella*) that likely reflects the natural resistome of bat intestinal microbiota.

In contrast, anthropogenic ARG inputs at downstream water sites are characterized by aminoglycoside resistance genes (APH, AAC(6')), tetracycline and macrolide resistance, and notably the *sul1-qacEΔ1* integron cassette. This cassette is a well-established molecular marker of anthropogenic contamination, commonly associated with human wastewater, hospital effluent, and agricultural runoff (Haenelt et al., 2023; Gillings et al., 2015). The Mungyeong study site is a remote upland karst area with no wastewater treatment plant, hospital, or large-scale agricultural facility in the catchment; the only nearby anthropogenic input is a small residential cluster ~600 m downstream of the caves at W8 (Fig. 1). Its dramatic elevation at W8 (*sul1*: 211,926 RPKM, 4.4-fold higher than any other site) provides strong evidence for wastewater or agricultural input near the residential area. Consistent with this, MST analysis identified W8 as the only water site with substantial human fecal contribution (5.9% in August), accompanied by pig (1.6%) — a fecal profile entirely absent at upstream cave sites. The co-occurrence of human fecal markers from MST and anthropogenic ARG markers from metagenomics at the same site provides independent, cross-methodological confirmation of domestic wastewater input.

The convergence of these two distinct ARG sources—a wildlife-derived reservoir from cave interiors and anthropogenic inputs at downstream residential sites—creates a mixing zone where resistance determinants of different ecological origins co-occur within the same watershed over a distance of only ~600 m. This spatial compactness amplifies the public health risk: communities consuming water from downstream Yaksu facilities are simultaneously exposed to bat-derived enteric pathogens and anthropogenic antibiotic resistance determinants. Interventions targeting only one source (e.g., restricting cave access without addressing downstream wastewater) would be insufficient to mitigate the full spectrum of ARG exposure.

4.4. Co-localized ARGs and VFs in clinically significant pathogens

Perhaps the most concerning finding is the identification of contigs simultaneously carrying ARGs and VFs, corresponding to clinically significant pathogens. The UPEC contigs from guano carry multidrug efflux systems co-localized with iron acquisition operons (*chu*, enterobactin) — a combination characteristic of successful uropathogenic lineages that cause urinary tract infections. *S. enterica* Typhimurium contigs co-carry efflux pumps with curli fimbriae for biofilm formation — a major foodborne pathogen with enhanced environmental persistence (Castelijn et al., 2012; Yaron and Romling, 2014; Zhuo et al., 2024). *S. dysenteriae* contigs carry TolC efflux with a complete Type II secretion system — the causative agent of bacillary dysentery.

The chromosomal encoding of most ARG-VF contigs (non-mobilizable by plasmid analysis) suggests stable inheritance, meaning resistance traits are unlikely to be lost in the absence of antibiotic selection. The detection of multidrug-resistant *P. aeruginosa* with motility capabilities in cave stream water (W1) further indicates that the aquatic environment sustains clinically relevant opportunistic pathogens equipped with both resistance and virulence capabilities.

4.5. One health implications

This study reveals a system in which all three pillars of the One Health framework converge within a single watershed: wildlife (bat guano as a reservoir of ARG-carrying enteric pathogens), the environment (karst hydrology facilitating microbial transport from cave interiors to surface streams), and human activity (anthropogenic ARG inputs and direct consumption of untreated spring water). The downstream mixing zone, where wildlife-derived and anthropogenic ARG signatures co-occur, exemplifies the type of interface that One Health frameworks are designed to address—one in which risks cannot be attributed to or managed through any single sector alone.

Importantly, the identification of bat guano as a wildlife-derived ARG reservoir should not be interpreted as justification for habitat disturbance or population control of bat colonies. Bats provide critical ecosystem services including pest suppression, pollination, and seed dispersal, and many Korean cave-roosting species are legally protected as endangered wildlife. Rather, these findings provide a basis for conservation-informed management options to be evaluated through stakeholder engagement and risk-benefit analysis: enhanced molecular water-quality monitoring at downstream Yaksu facilities, summer-period advisory systems coinciding with bat breeding and monsoon flushing, restricting human access to roosting caves during peak activity, and where feasible, physically separating water intake points from guano-contaminated flow paths.

4.6. Limitations

Several limitations warrant consideration. First, metagenomic and amplicon-based methods detect DNA regardless of cell viability, so a follow-up validation plan combining propidium-monoazide (PMA)-qPCR targeting priority pathogens (UPEC, *Salmonella*, *Shigella*, *P. aeruginosa*) at W1 and W8, selective enrichment culture with antimicrobial susceptibility profiling, and whole-genome sequencing of cultured isolates would confirm the multidrug-resistant genotypes inferred from metagenomic assembly.

Second, while the *sul1-qacEΔ1* integron cassette strongly suggests human wastewater input at downstream sites, direct sampling of wastewater and agricultural runoff in the W8 catchment, paired with host-marker qPCR, would provide ground-truth confirmation; and (c) formal quantitative microbial risk assessment (QMRA) is needed to translate detection data into actionable risk thresholds for public water use. Sampling across only two months (July and August) limits our ability to distinguish seasonal effects from stochastic events; long-term monitoring across a full annual cycle, including winter (when many Korean bat species hibernate and guano deposition is expected to be minimal), is needed before generalizing the temporal pattern.

Third, both ARG quantification and pathogen screening have inherent resolution constraints. Read-based RPKM normalization of ARG abundance is subject to coverage bias from contig assembly depth, an acknowledged limitation of read-based ARG quantification (Bengtsson-Palme, 2018) that we mitigate by additionally reporting assembly-based ABRicate annotations as an orthogonal high-confidence call; 16S rRNA amplicon-based pathogen screening operates at genus-level resolution, whereas pathogenicity is often species- or strain-specific, and full-length 16S sequencing on long-read platforms such as PacBio, which the pipeline already supports, would provide species-level resolution.

Fourth, the source reference database is dominated by Korean fecal samples. Although the Australian generalizability test demonstrated reliable cross-region transfer for cow, horse, and chicken, human attribution was qualitative only at low spike levels in the in-silico mocks, reflecting known geographic divergence of human gut microbiomes; users in non-Korean settings are encouraged to augment the bundled database with locally collected reference samples. Our cross-platform PacBio HiFi validation similarly used only three Italian human fecal samples: the V4 sub-region extraction is host-agnostic by design and our companion paper (Unno, 2026) demonstrates HiFi extraction on a non-host environmental dataset (soil), but HiFi 16S data from non-human animal sources were not available because the platform is relatively new. Finally, larger and more geographically diverse wild-bird references would improve resolution for systems where avian inputs are expected to dominate.

5. Conclusions

This study developed and validated MST-Pipeline, an automated web-based platform for microbial source tracking with cross-region

harmonization, and applied it alongside shotgun metagenomics to investigate bat guano contamination of karst spring water in the Mun-gyeong region of Korea. The pipeline attributed fecal contamination sources with dose-dependent responses and cross-platform compatibility. Source tracking identified bat guano as the highest-contributing fecal source among the nine host groups tested (including a wild-bird reference), with contributions higher in August (mean 4.76%) than July (1.31%), driven by increased bat activity and antecedent rainfall; bird contribution remained negligible across all samples (mean 0.32%, bat-to-bird ratio 9.6:1), confirming that the bat-dominance result is robust to inclusion of avian sources. The downstream residential site showed a distinct anthropogenic fecal signature with human and pig inputs. Metagenomic analysis revealed dual ARG sources within the watershed: a wildlife-derived reservoir in guano dominated by enteric efflux systems and fluoroquinolone resistance, and anthropogenic inputs downstream characterized by aminoglycoside resistance and *sul1-qacEΔ1* integron cassettes. Genome-resolved analysis identified multidrug-resistant UPEC, *S. enterica* Typhimurium, *S. dysenteriae*, and *P. aeruginosa* carrying co-localized ARGs and virulence factors on the same contigs, with chromosomal encoding suggesting stable inheritance.

Although fecal contributions estimated by source tracking are low (1–4%), they should not be interpreted as evidence of clean water; complementary shotgun metagenomics confirmed the presence of antibiotic-resistant pathogens carrying co-localized ARGs and virulence factors in the same samples. This underscores the need for routine molecular monitoring of water sources conventionally regarded as clean. The perception of karst spring water as naturally pure must be reconsidered in light of the antibiotic-resistant pathogenic bacteria it carries. These findings provide a scientific basis for implementing seasonally informed monitoring, conservation-compatible risk mitigation, and integrated One Health surveillance in regions where wildlife reservoirs overlap with human water resources.

Data availability statement

The karst water amplicon and shotgun metagenomic sequencing data generated in this study will be deposited in the NCBI Sequence Read Archive upon acceptance. The spike-in validation data are available under BioProject PRJNA1417490. Bat fecal microbiome reference data are available under BioProject PRJNA1436137. Livestock, human, and environmental reference data are available under BioProjects PRJNA1071275 and PRJNA10195. MST-Pipeline source code is freely available at <https://github.com/tatsu1207/MST-PipelineundertheMITLicense>.

Declaration of generative AI use

During the preparation of this work the authors used Claude (Anthropic) in order to assist with data analysis scripting and manuscript drafting. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Tatsuya Unno: Writing – original draft, Supervision, Software, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Geon Choi:** Investigation. **Jae-Hyeon Oh:** Resources, Formal analysis, Data curation. **Jun Heo:** Resources, Investigation. **Dukki Han:** Resources, Investigation. **Jeonghwan Jang:** Writing – review & editing, Conceptualization. **Soyeon Park:** Resources, Project administration, Methodology, Investigation. **Jae-Yeon Kang:** Methodology, Investigation. **Jangwon Seo:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.wroa.2026.100582](https://doi.org/10.1016/j.wroa.2026.100582).

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